

JX-2R-01LLA

Thermal Printer Mechanism Specification

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1. Introduction

1.1 JX-2R-01 LLA Micro Thermal Printer Module

Several patent designs of Jingxin Company make the printing more uniform and the operation more stable. The movement has the characteristics of small size, ultra-low working voltage (3.0 ~ 5.5 DC V), high efficiency, etc. The unique design makes the JX-2R-01 LLA movement have the characteristics of low working voltage, easy paper loading, low noise and high reliability.

1.2 Characteristics

- ◆ Easy-load paper
- ◆ Small size and light weight
- ◆ High temperature resistant plastic base, stable, reliable, long life, excellent heat dissipation performance
- ◆ Fast printing speed: up to 70mm/s (motor drive voltage is 5V DCV; 2-2-phase drive)
- ◆ Low working power supply (working power supply voltage V_H of TPH: 3.0 ~ 5.5 DC V)
- ◆ High printing precision (horizontal 8 dots/mm, vertical 16 dots/mm — motor 2-2-phase drive)
- ◆ Long service life of mechanical wear resistance (more than 50km)
- ◆ Low noise: brushless magnetic excitation stepper motor drive; high wear resistance, high and low temperature resistant special engineering plastic gear composition, so that the transmission noise is extremely low.
- ◆ Applicability: JX-2R-01LLA is applicable to miniature thermal bill printers with ultra-low working voltage and UnionPay handheld thermal bill printers (powered by a single battery) (special note: the power supply

voltage to the print head should not be higher than 5.5V, otherwise the print head will be damaged).

1.3 Description

This manual describes the electrical and mechanical characteristics, quality assurance and maintenance of JX-2R-01LLA core, namely, the composition mechanism, basic parameters, scope of application, definition and structural dimensions of peripheral interfaces, as well as precautions and maintenance details in use. Our company reserves the right to amend and improve this manual. If you need the latest version of this manual, you can directly contact Xiamen Jingxin Technology Co., Ltd; Jingxin Technology is currently composed of Xiamen Jingxin Technology Co., Ltd. and Ningbo Jingxin Technology Co., Ltd. Xiamen Jingxin Technology Co., Ltd. is positioned as the product research and development, quality system and after-sales service center of Jingxin Technology Group, while Ningbo Jingxin Technology Co., Ltd. is positioned as the mass production base and sales center of Jingxin Technology Group. (Xiamen, Ningbo) Jingxin Technology Co., Ltd. does not guarantee the reliability of the work if the design is not in accordance with this manual.

2. Specifications and parameters

Table 1

Project	Explanation
Print method	Heat-sensitive
Effective Print Width (mm)	48
Print dot density resolution (dots/mm)	8 (landscape), 16 (portrait)
Print the number of points	384 points/line
Paper width (mm)	58MAX
Dot Spacing (mm)	0.125
Point size	0.125mm (transverse) X0.0625mm (longitudinal)
Maximum print speed	70mm/s (DC 5V for motor drive voltage)
Paper feeding accuracy	0.0625mm (one step distance, 2-2phase drive)/0.03125mm (one step distance, 1-2phase drive)
Print head temperature detection	Thermistor
Paper out detection	Reflected light sensor
Print head operating voltage (DCV)	3.0~5.5
Logic operating voltage (DCV)	3.3(3.0~4.5)
Motor operating voltage (DCV)	3.2~5.5
Operating temperature	-5 °C ~ + 50 °C (no condensation)
Operating humidity	10% ~ 90% RH (no condensation)
Storage temperature	-40 °C ~ + 80 °C (no condensation)
Storage humidity	5% ~ 90% RH (no condensation)

Mechanical noise	Less than 60dB (A-weighted effective value)
Number of opening and closing of rubber roller	More than 5000 times (one time for the rubber roller to leave and reset)
Traction force to thermal sensitive paper	≥50g
Grip brake force for thermal paper	≥80g
Working life	The wear resistance of the mechanism and the print head is > 50km, and the electrical life of the print head is 108 pulses (in the rated state)
Weight (G)	40.5g
Dimensions (L X W X H)	70±0.2mm*33±0.2mm*15.3±0.2mm
Suitable for thermal printing paper	Conventional base stock (Oji, Mitsubishi, Ricoh), thickness 0.06 to 0.08 mm (density — 60 to 90 grams) — conventional thermal properties;

3. Print head parameters

3.1 Rated parameters

Table 2

Heating points	384 points/line
Dot pitch	0.125mm
Point size	0.125mm (horizontal) X 0.0625mm (vertical)
Printable width	48 mm
Overall width	54 mm
Average resistance	90Ω±5%
Operating voltage	3.0~5.5 DC V
Pulse life	10 ⁸ pulses
Mechanical wear life	50km
Life test strip conditions	25℃
The proportion of heating time is not more than	12.5%.

3.2 Maximum allowable parameters (ambient temperature: 25 °C)

Table 3

Parameter	Code name	Maximum rating	Conditions
Heating Energy	E0	0.25mJ/dot	Tsub=25°C
Heating voltage	VH	5.5 V	
Logic voltage		5.5V	Including the highest voltage
Maximum number of simultaneous working points	Ndot max	96 dots	
Operating temperature	Ta	-5~50°C	The detection temperature of the thermistor shall not exceed 70 °C, and it is recommended that the recovery printing temperature be lower than 60 °C.

3.3. recommended parameters :

Table 4

Project		Code name	Electrical parameters	Conditions
Power consumption		P_o	0.094W/dot	Rav = 90Ω, while printing: 96 dots
Supply voltage		VH	3.3V	
Cycle printing		S.L.T	2.5ms/line	
Cons umpt ion Ener gy	5℃	E_o (Ton)	0.16mJ/dot(1.71ms)	
	25℃		0.14mJ/dot(1.49ms)	
	45℃		0.12mJ/dot(1.28ms)	
Current consumption		I_o	32.3 mA/dot	

3.4 Calculation formula

The heating energy can be calculated by the following formula:

$$E_0 = I_0^2 \overline{Rt_s} = \frac{(VH - V_{com})^2 \bullet \overline{R} \bullet t_s}{(R + R_{ic})^2}$$

$R_{ic} = 3\Omega$: internal resistance of driver IC

t_s : Gating pulse width (heating time)

VH : Print voltage

$\overline{R}$: Resistance value of heating element

$V_{com} = 0.3V$: common electrode voltage drop

3.5 Thermistor

Calculation formula:

(Note: T: absolute temperature) $R_x = R_{25} \times EXP[B \times (\frac{1}{T_x} - \frac{1}{T_{25}})]$

Among

B constant 3950 K $\pm$ 3%

R25 Resistance value 30 K Ω $\pm$ 5% (at 25 $^{\circ}$ C)

Tx operating temperature -20 $^{\circ}$ C $\sim$ + 80 $^{\circ}$ C

T25 Operating temperature of 1 punch width 25 $^{\circ}$ C Thermistor thermometer

温度 ($^{\circ}$ C)	热敏电阻阻值 (R)		
	Min. (k Ω)	Typ. (k Ω)	Max. (k Ω)
-40.0	717	843	989
-35.0	535	623	723
-30.0	405	466	535
-25.0	308	352	400
-20.0	238	269	303
-15.0	185	208	232
-10.0	145	161	178
-5.0	113	124	137
0.0	88.7	96.8	105
5.0	69.9	75.7	81.7
10.0	55.4	59.5	63.8
15.0	44.1	47.1	50.1
20.0	35.4	37.5	39.6
25.0	28.5	30.0	31.5
30.0	22.8	24.2	25.5
35.0	18.3	19.6	20.8
40.0	14.9	15.9	17.1
45.0	12.1	13.1	14.1
50.0	9.92	10.8	11.7
55.0	8.16	8.91	9.7
60.0	6.76	7.41	8.12
65.0	5.62	6.2	6.83
70.0	4.7	5.21	5.77
75.0	3.95	4.4	4.9
80.0	3.34	3.74	4.18

3.6 Electrical parameters (25 $\pm$ 10 $^{\circ}$ C)

3.6.1. See Table 5 below for electrical parameters when Vdd is 5V:

Table 5 (Vdd is 5V)

Project	Code name	Minimal	Typical	Max	Unit	Conditions
Print voltage	VH	--	3.3	5.5	V	
Logic voltage	Vdd	3.0	3.3	4.5	V	
Input voltage (high)	VIH	0.7Vdd	--	Vdd	V	
Input voltage (low)	VIL	0	--	0.3Vdd	V	
Latch Input Current (High)	IIH	--	--	3.0	μA	VDD=3.0V VIH =3.0V
Heating input current (high)		--	--	300		
Clock Input Current (High)		--	--	3.0		
Data Input Current (High)		--	--	0.5		
Latched Input Current (Low)	IIL	-300	--	--	μA	VDD=3.0V VIL =0V
Heating input current (low)		-3.0	--	--		
Clock Input Current (Low)		-3.0	--	--		
Data Input Current (Low)		-0.5	--	--		
Clock frequency	fclk	-	-	10	MHz	
Clock pulse width	tw(T)	20	-	-	ns	
Data setup time	tsu(D)	20	-	-	ns	
Data hold time	th(D)	20	-	-	ns	
Latch set time	tsu(LA)	20	-	-	ns	
Latch pulse width	tw(LA)	20	-	-	ns	
Latch to strobe set times	tsu(STB)	320	-	-	ns	
Gate-to-Latch Setup Time	th(STB)	36	-	-	μs	
Clock to Data Output Delay Time	td(SO)	-	-	80	ns	
Gate-to-Drive Output Delay Time	td(DO)r	-	12	36	μs	
	td(DO)f	-	12	36	μs	
Consumed current	IDD	-	8	12	mA	VDD=3.3V fCLK =10MHz DI = 1/2 fCLK

3.6.2. The timing architecture is shown in Figure 1 below:

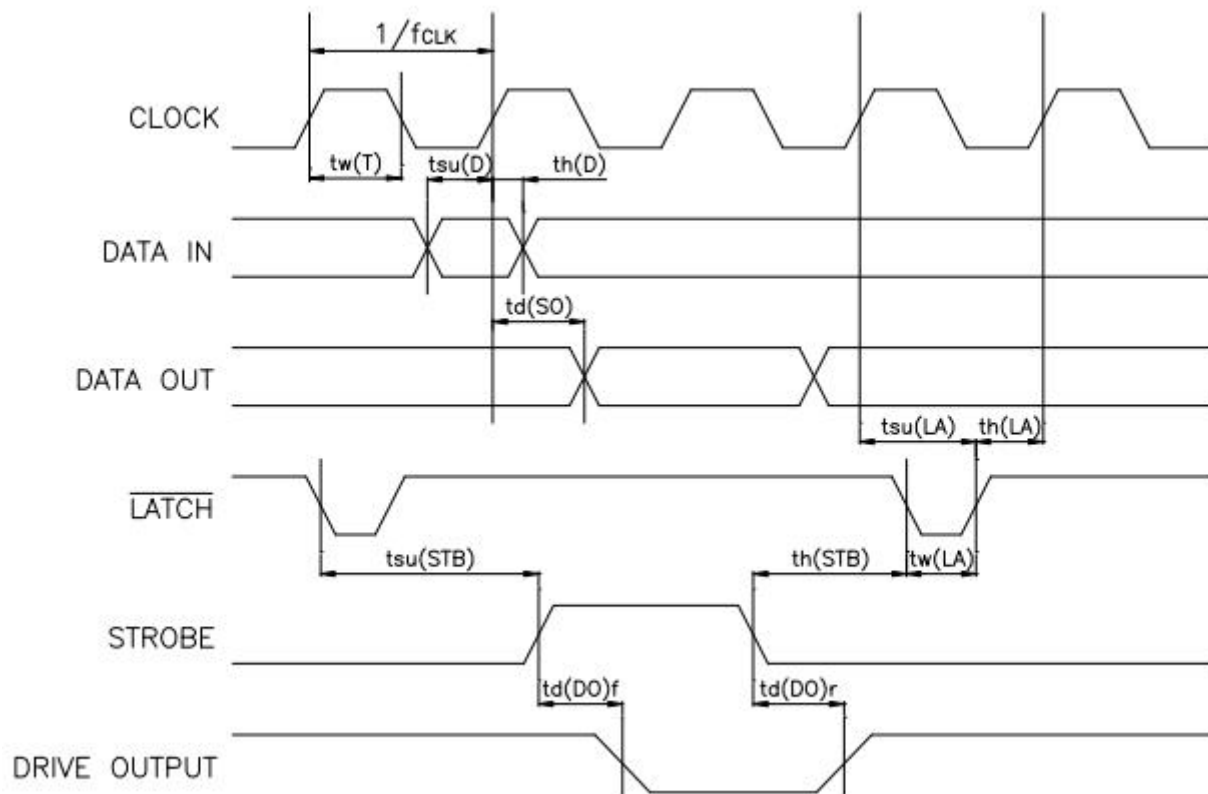


Figure 1

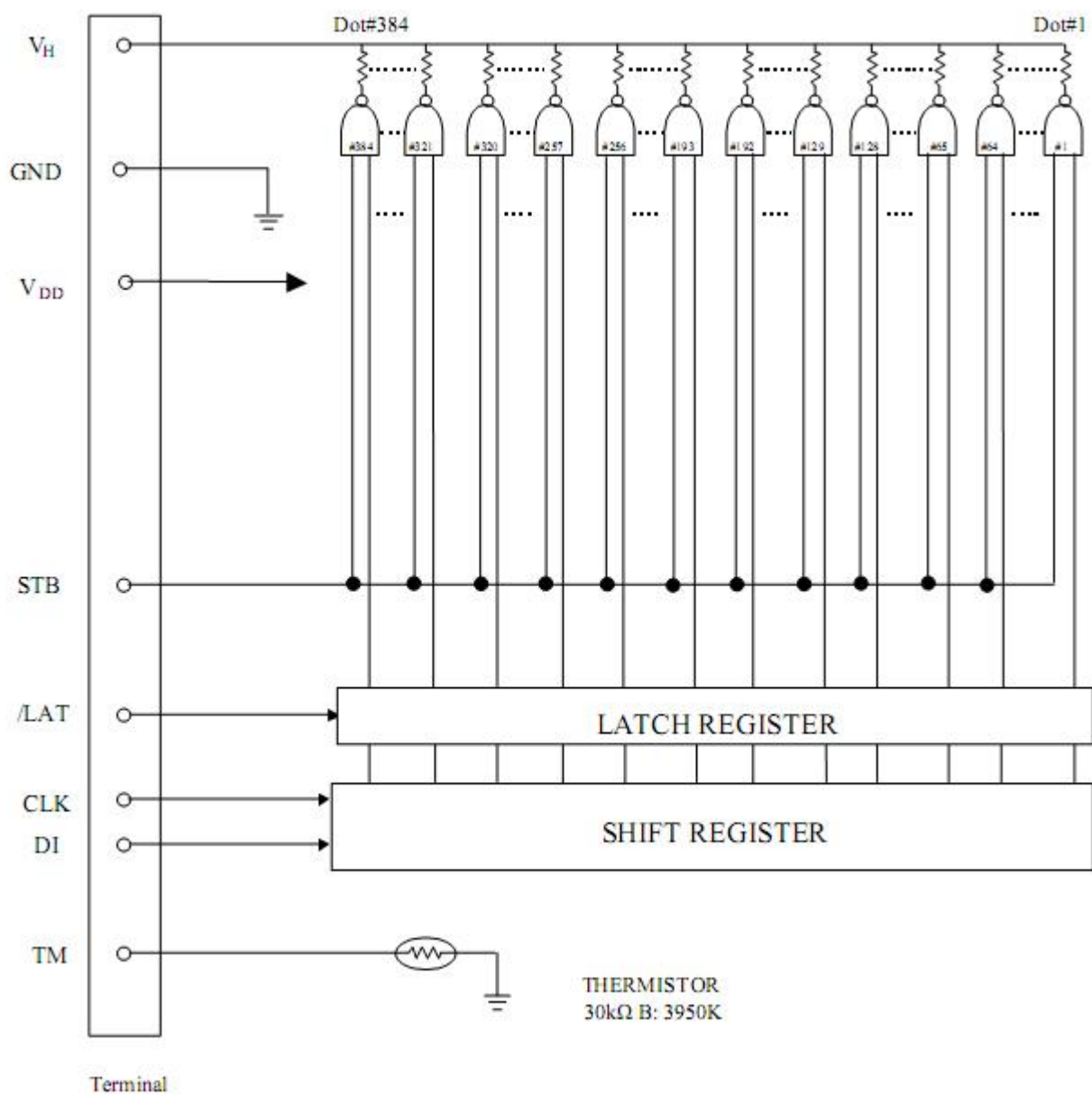


Figure 2

Table 6

Strobe (STB) Number	Point number	Point/foot
1	1 to 384	384

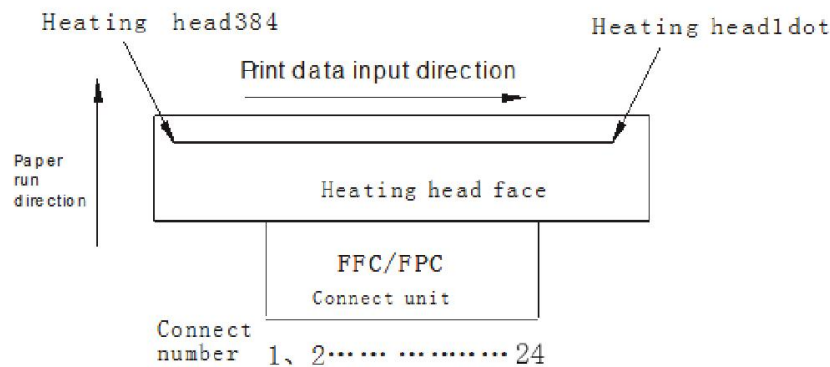


Figure 3

(Note: This is a directional view of the print line facing the print head.)

3.7 Cautions

Electrical attention

1) When the printing device is in the standby state (or non-printing state), the printing power (VH) must be turned off.

2) The power supply of the print head shall be in the following order:

Turn on sequence: logic power supply (VDD) first, then print power supply (VH).

Shutdown sequence: print supply (VH) before logic supply (VDD).

3) The STROBEs signal should remain in the "disable" state when the print head power is ON/OFF.

4) If the applied voltage exceeds the limit range of the driver IC or in the case of instantaneous high voltage, the lock effect may damage the print head, so the power supply and signal voltage (including the oscillation peak) should be controlled within the following range:

VH : 0~7.0DC V

VDD : 0~6.0V

Other signals: GND ~ VDD + 0.3V

5) Logic signal input is CMOS level (e.g. 74HC 240 or equivalent).

6) In order to prevent the driving chip and heating element from being damaged by static electricity, do not touch the interface terminal directly with your hand, and do not let the terminal touch the charged

object. It is recommended to install a brush beside the thermal print head when assembling the thermal print head.

7) In order to keep the power supply stable and prevent instantaneous high voltage, it is recommended to add electrolytic or ceramic capacitors between the printing power supply (VH) and the logic power supply (VDD) and the ground wire.

8) The length of the cable connected to the print head should be less than 30cm. It is recommended to use a flat cable with a protective layer.

Mechanical attention

1) The surface of the print head substrate should avoid heavy pressure and scraping. The heating element is easy to break when it is mechanically impacted, so it should be prevented from touching any hard objects.

2) Avoid condensation (condensation) on the print head surface. If condensation occurs, do not turn on the power of the print head until the condensation disappears.

3) Avoid using inferior paper or other media with rough surface, otherwise the life of the print head will be reduced. Therefore, please confirm and evaluate the quality of the thermal medium before use.

4) The flatness of the print head monomer has been controlled to the minimum with the change of temperature. When installing the fixed print head, please avoid distortion or deformation.

5) The phenomenon of paperless printing should be avoided, otherwise the resistance of the heating element will be damaged.

6) The thermal print head is a heating device, so if there is a mechanical or electrical abnormality, the print head may cause damage to the heating element or smoke and fire due to excessive heating.

Therefore, please confirm that the thermistor of the print head is used to control the temperature of the print head, and the corresponding power cut-off system is designed to protect the print head in case of abnormality.

7) Na⁺, K⁺ and Cl⁻ ions will corrode and damage the print head. Please confirm and evaluate the

reliability of the components that are often in contact with the print head, such as rubber roller, thermal paper and other consumables.

Operation attention

1) When there are paper scraps, dust or ink residues on the heating element, please use flexible materials dipped in alcohol or detergent to wipe it gently. It is strictly prohibited to use hard materials such as sandpaper. The electrical test can be carried out only after the alcohol on the surface of the printing head is volatilized.

2) The interface socket or terminal is easy to bend, so it shall not be damaged during plugging and unplugging.

Shelf life

1) The shelf life of the product is one year. Within 12 months from the date of delivery of the products, the quality of raw materials or processing of the products shall be guaranteed within the scope of limited liability.

2) After SHEC analysis, it is determined that the defective products are caused by the above reasons, and the defective products will be repaired or replaced.

8. FPC/FFC hot pressing:

In order to avoid defects such as package glue crack and gold wire breakage caused by internal peeling of non-glue crack which may occur during hot pressing of FPC,

SHEC shall pack the product with nitrogen. However, it is also necessary to carry out the management of dampness and actual installation conditions after opening.

Your attention is drawn to the following points:

1) Carry out FPC/FFC hot pressing as soon as possible after the product is unsealed. If the FPC/FFC is not hot-pressed after unsealing, please add desiccant to seal the opening. (Such as tape seal after folding)

2) Please control the temperature, humidity and storage period of the product storage place to prevent the product from getting damp. (MSL-4 grade is recommended)

3) It is recommended to dry and dehumidify the product before hot pressing FPC/FFC.

4) Strictly control the peak temperature during hot pressing, and operate according to the standard of the best quality condition verified.

4. Stepping motor

The paper advances 0.03125mm (1-2-phase drive)/0.0625mm (2-2-phase drive) per step of the stepper motor.

4.1 Stepping motor parameters

Table 7

Project	Specifications	Conditions
Rated voltage	3.2~5.5 DCV	
Phase	Phase 2	
Step Angle	18 ° (2-2 phase excitation)/9 ° (1-2 phase excitation)	
Step distance	0.0625 mm	2-2-phase excitation
Phase resistance	10 $\Omega \pm 7\%$	20℃
Phase current	0.40A	
Drive mode	Bipolar, 1-2-phase drive	

4.2 Phase of stepping motor

The stepper motor model JX-2R-01LLA adopts 1-2 phase driving mode, with 4 positions for advancing one dot matrix line, as shown in the table below:

Table 8

Location	STEP1	STEP2	STEP3	STEP4	STE P5	STE P6	STE P7	STE P8
A	+	-	-	-	-	-	+	+
B	-	-	-	-	+	+	+	-
$\bar{A}$	-	-	+	+	+	-	-	-
$\bar{B}$	+	+	+	-	-	-	-	-

Driving procedures of the stepping motor (1-2phase)

Motor shift in the CCW (seen from motor gear side)

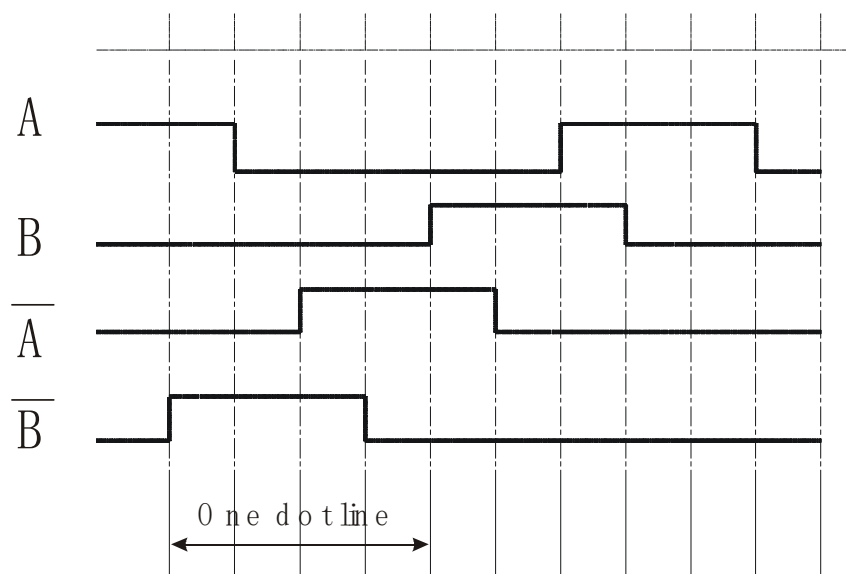


Figure 4

2-2-phase time sequence table

Location	STEP1	STEP2	STEP3	STEP4
A	+	+	-	-
B	+	-	-	+
$\overline{A}$	-	-	+	+
$\overline{B}$	-	+	+	-

1-2-phase motor drive pulse frequency: during the paper feeding process, the drive speed of the motor shall be lower than the value obtained from the equation.

$$P_m = V_p \times 960 - 1120 \text{ (pps)}$$

Maximum motor drive pulse rate at P_m : V_p

V_p : Motor drive voltage

Reference voltage maximum driving pulse meter

Voltage	Maximum pulse
3.0V	1760 pps
4.0V	2720 pps
4.2V	2720 pps

The motor speed should be less than 960 PPS during the backward feed.

4.3. Stepping motor drive

It is recommended to use PWM mode to drive the stepper motor, such as L3967, LB1836, BD6735 and other driver chips. Use different drive currents for different motor speeds. In this way, the heating of the stepping motor can be effectively reduced, and the noise during stepping printing can be effectively reduced.

In order to avoid overheating of the stepping motor, the duty cycle lower than that in the above table shall be adopted as far as possible in the design of

the whole machine; and the longest power-on time shall not exceed 30 seconds.

5. Paper shortage detection

5.1、Paper shortage detection

JX-2R-01 LLA movement adopts a reflective photoelectric on-off detection sensor. The main functions of the photoelectric detection sensor are as follows:

A: Paper shortage detection

B: The printing paper can also be positioned by the marks on the printing paper.

As shown in the figure below, when there is no paper, the light emitted by the photoelectric detector cannot be reflected, and the output is high. When the paper is normal, the light emitted by the photoelectric detector is reflected and received by the receiving tube, the output is low level, and the circuit of the photoelectric switch is driven, as shown in the figure below. When the paper is out of paper, do not start the printer to heat. When there is no paper or before the paper is led out of the print head by the automatic paper feed, the paper feed speed must be reduced (reduce the drive pulse frequency of the drive motor: generally 400 ~ 600PPS).

Table 9 Maximum Parameter Values

Project	Code name	Numerical value	Unit
---------	--------------	--------------------	------

Input	Forward current	IF	50	mA
	Reverse voltage	VR	5	V
	Minimum power	P	70	mW
Output	Collector to emitter voltage	VCEO	20	V
	Emitter-collector voltage	ECO	5	V
	Collector current	IC	20	mA
	Minimum collector power	PC	70	mW

Table 10 Photoelectric parameters (25 °C):

Project		Code name	Conditions	Numerical value			Unit
Input end	Forward voltage	VF	IF=10mA	1.0	1.2	1.6	V
	Reverse current	IR	VR=5V			10	μA
Output terminal	Collector to emitter breakdown voltage	BVCE0	IC=0.5mA	20			V
	Emitter-collector breakdown voltage	BVECO	IE=0.1mA	5			V
	Collector dark current	ICEO	VCE=10V IF=0mA			200	nA
Coupling characteristics	Collector to emitter saturation voltage drop	VCE(SAT)	IC=2mA Ee=1mW/cm2			0.4	V
	Sensor current	Ic	VCE=5V IF=10mA	150		600	μA

	Leakage current	ILEAK	IF=10mA VCE=5V No reflecting surface			1	μA
	Rise/Fall Time	Tr/Tf	VCE=5V IF=1mA RL=100 Ω		5/5		μs

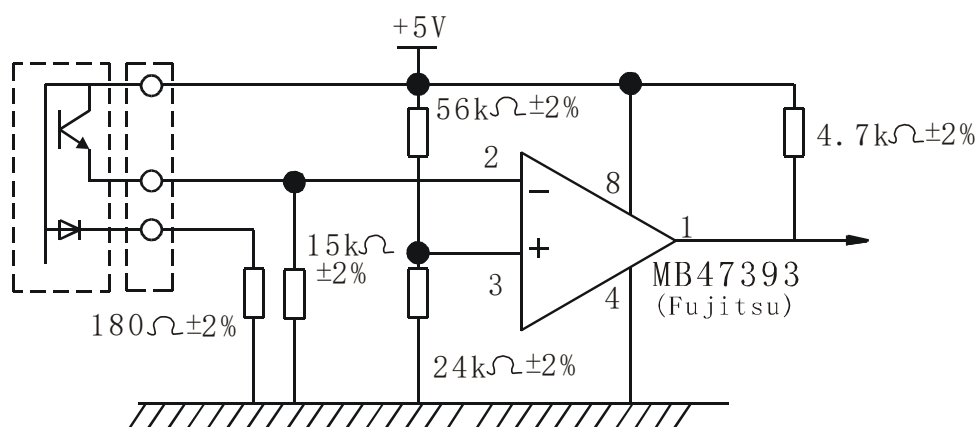


Fig. 5 Circuit diagram of photoelectric on-off detection sensor
(capacitance of 0.1 μF is added at the optocoupler end)

6. Interface definition

Table 11 Interface Definition Table

Number	Signal	Explanation	Number	Signal	Explanation
1	PHK	Cathode of Photo Interrupter Sensor	13	STB	Strobe pulse
2	VSEN	Power supply for photo interrupter sensor	14	TM	Thermistor
3	PHE	Emitter of photo interrupter sensor	15	GND	Place of printing
4	VH	Print head drive power supply	16	GND	Place of printing
5	VH	Print head drive power supply	17	GND	Place of printing
6	VH	Print head drive power supply	18	$\overline{\text{LAT}}$	Print data latch (memory store) signal
7	DI	Print data input	19	VH	Print head drive power supply
8	CLK	Print the clock	20	VH	Print head drive power supply
9	GND	Place of printing	21	A	Motor phase A
10	GND	Place of printing	22	$\bar{A}$	Motor phase $\bar{A}$
11	GND	Place of printing	23	B	Motor phase B
12	Vdd	Logic Power Terminal	24	$\bar{B}$	Motor phase $\bar{B}$

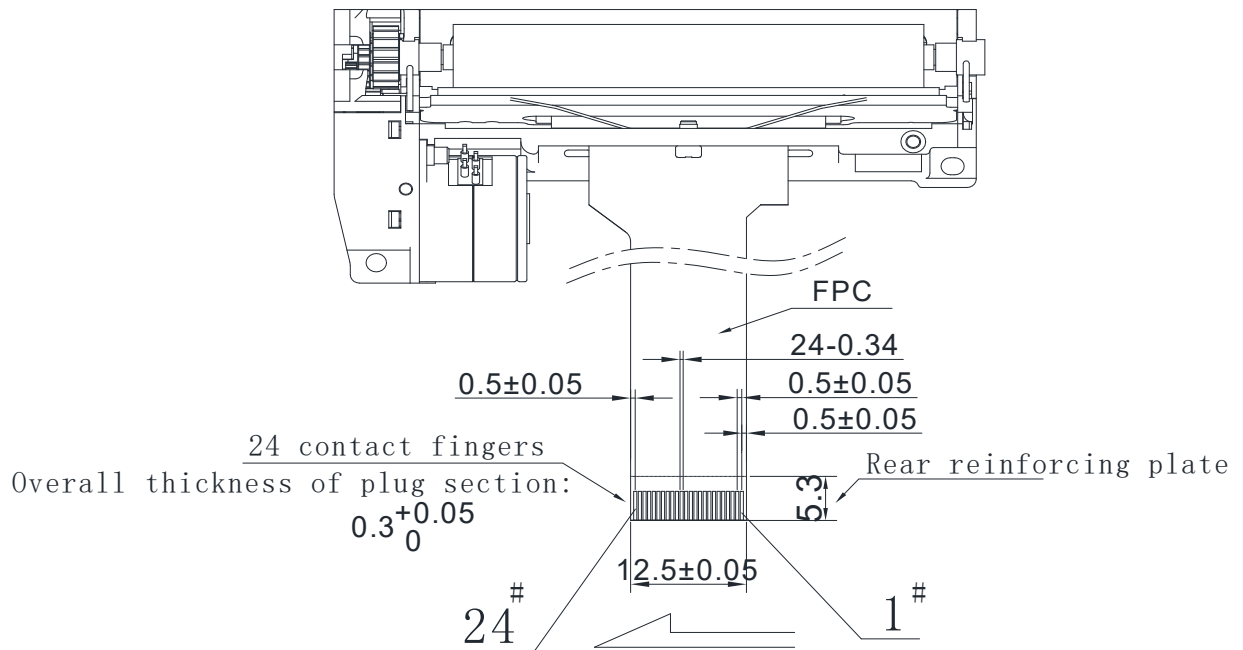


Figure 6 (Interface Definition Diagram)

7、Mechanical Design Reference

7.1、Pivot Design of Easy Paper Loading Mechanism

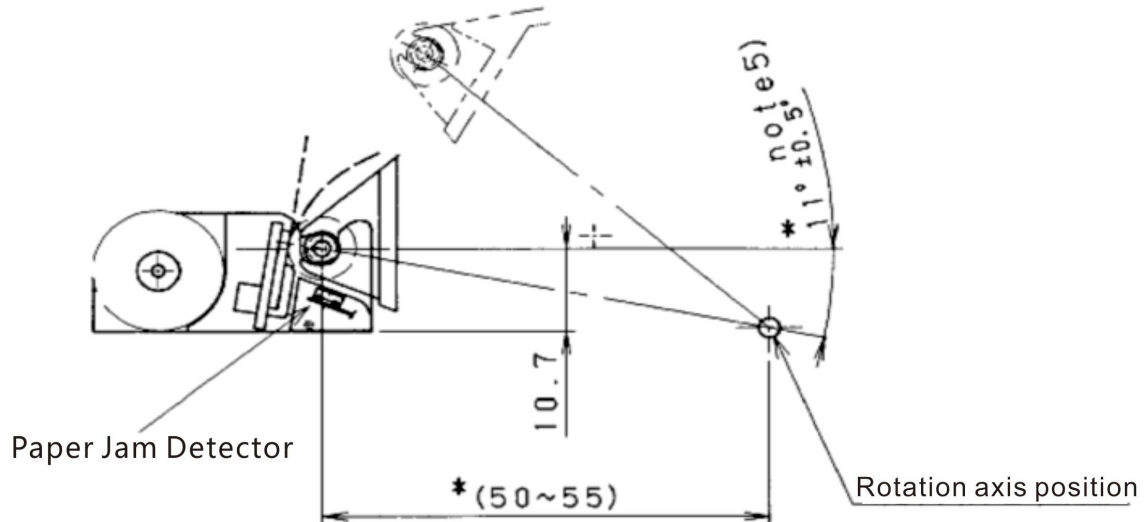


Figure 7

7.2、Structural Design of Paper Roll Position

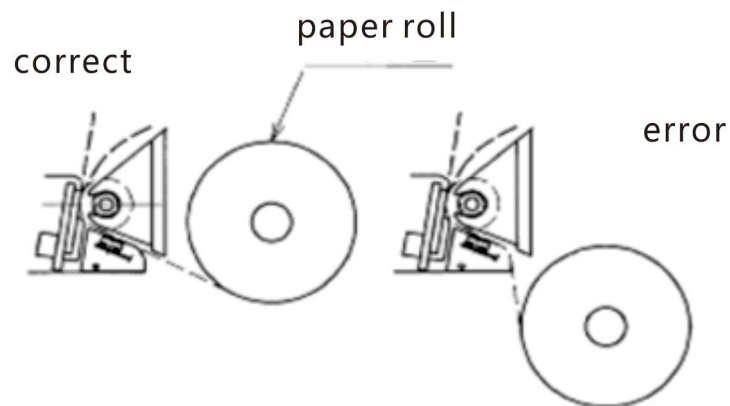


Figure 8

7.3、Assembly Position Dimension

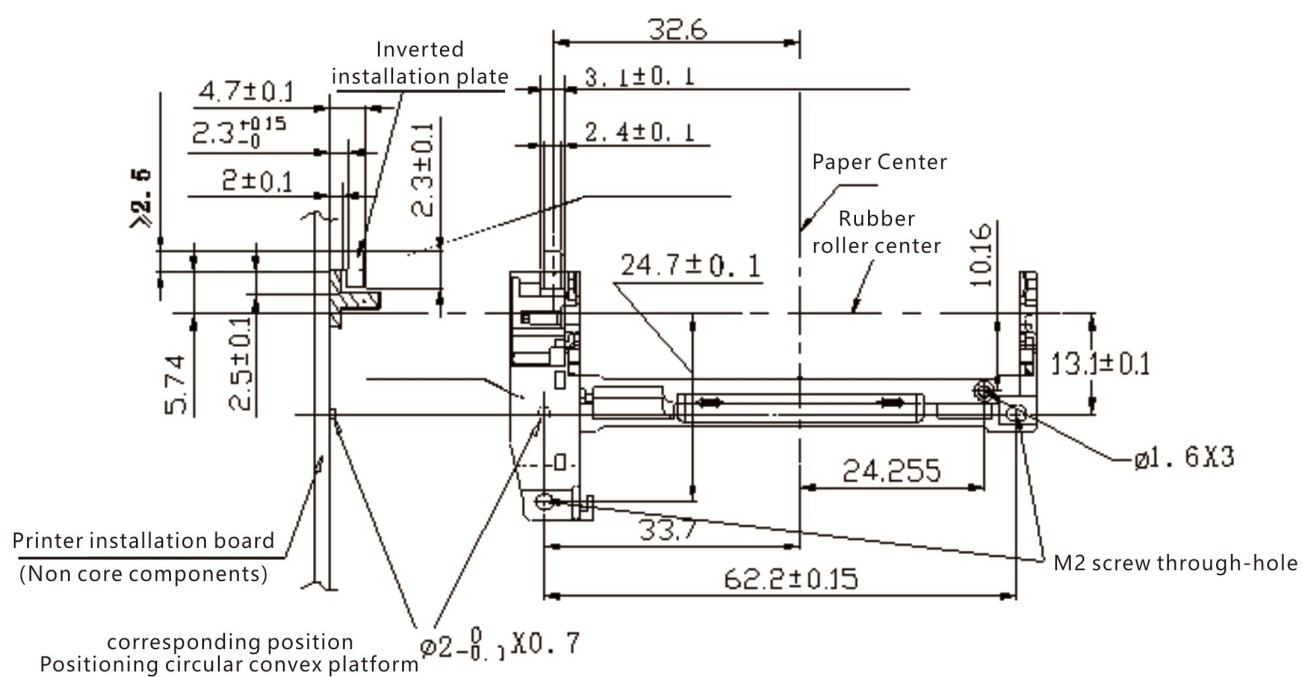
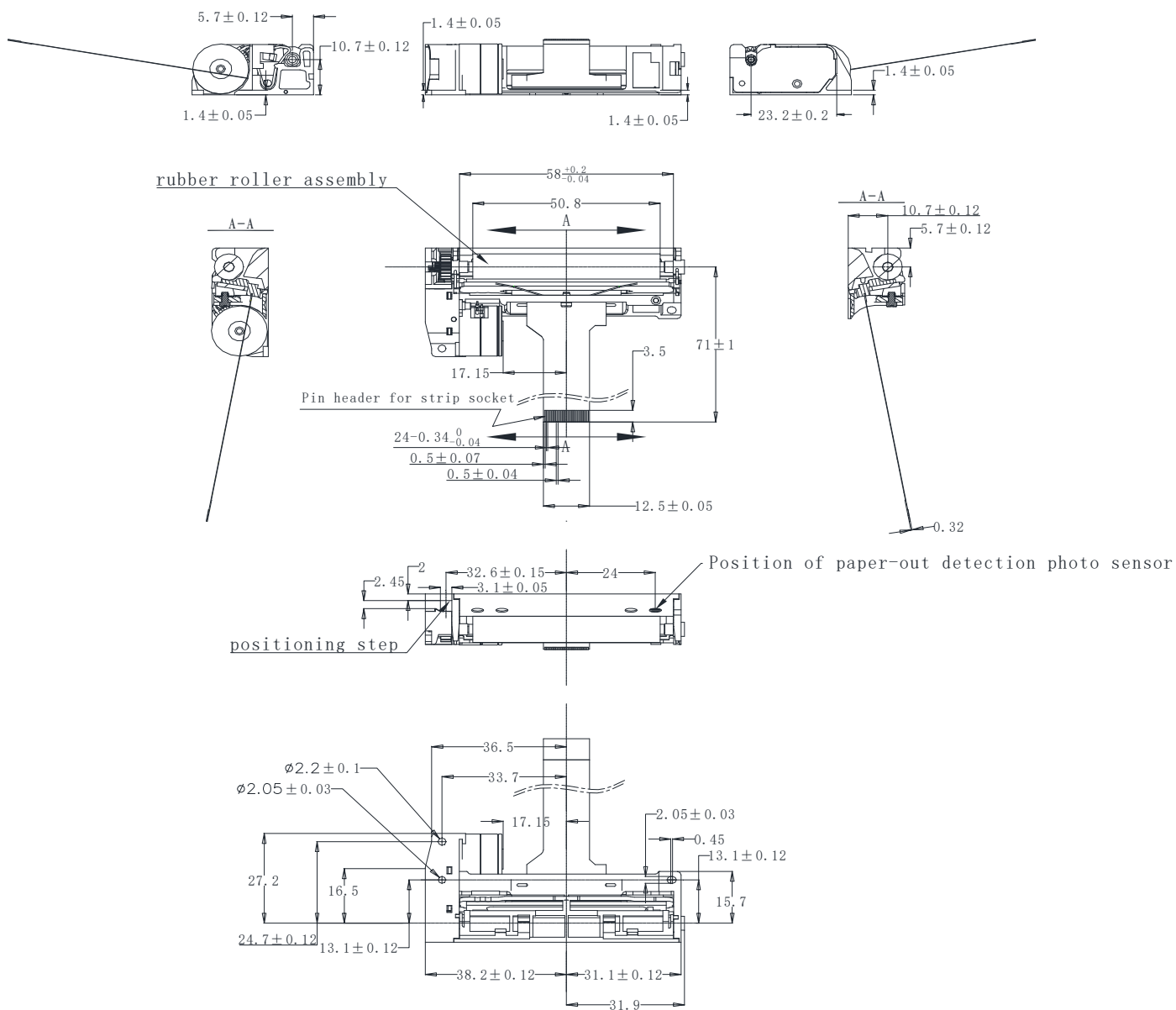


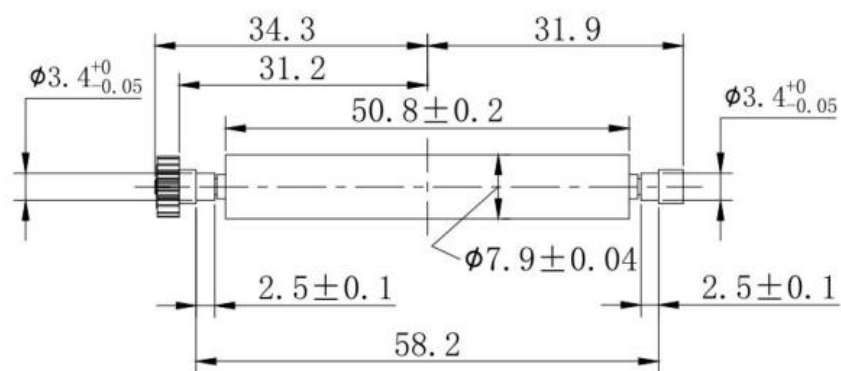
Figure 9

7.4、Appearance structure and size of the movement:



Unspecified dimensional tolerance: ±0.2 mm

Figure 10



Unspecified dimensional tolerance: ± 0.3 mm

Figure 11

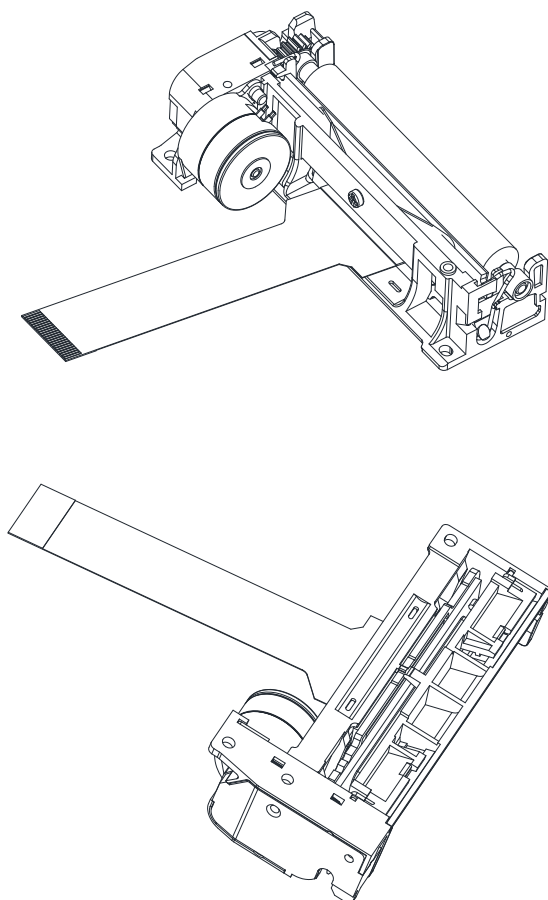


Figure 12

8. Precautions for movement application

- (1) The specific structure and appearance of the movement shall be subject to the official sample sent by Xiamen Jingxin Technology Co., Ltd.
- (2) Before testing and installing the complete machine, first confirm whether the model and specification on the movement are consistent with the requirements, and check whether they are the same as the samples at the time of identification.
- (3) Take care and handle with care during the inspection and maintenance of the movement and the production and assembly of the complete machine; pay attention to the anti-static measures for the working platform, assembly line, operators and turnover box.
- (4) The connection PIN end of the FPC of the core shall not be directly touched by the operator. When it is connected to the drive board of the complete machine, it shall be in a relatively loose state in space, and shall not be tensioned or subjected to additional force. When the FPC of the core is connected, it must be pulled and plugged when the power supply of the drive board of the core is reliably turned off; The FPC connected to the drive board shall not be plugged and unplugged for more than 10 times, and the FPC end connected to the print head in the core shall not be plugged and unplugged at will.
- (5) The storage and use of the movement should pay attention to waterproof, moisture, sunshine, corrosive chemical gases, etc. In the case of condensation of the movement, the power supply should be turned off immediately, and the movement can be used only after the water vapor is dried.
- (6) The control of the printing operation of the movement shall be carried out under the condition that there is paper. The forced operation without paper will cause the rubber roller rubber to directly rub the working and heating printing working surface of the TPH, which will greatly affect and shorten the service life of the TPH.
- (7) For other precautions, refer to the Precautions for Use of Thermal Printer Core prepared by Xiamen Jingxin Technology Co., Ltd.
- (8) Customers are advised to use thermal printing paper with good quality. Poor quality not only results in poor printing effect, but also increases the wear and tear on the print head of the movement, which will seriously shorten the service life of the print head.

9. Quality assurance and maintenance

The company guarantees that the product can give full play to its superior performance under normal use and maintenance, and the company also provides one-year warranty service.

9.1 Maintenance

9.1.1 The product shall be repaired by professionals with certain maintenance experience.

9.1.2 In case of maintenance, please do not replace the internal parts of the movement without authorization. If the user repairs blindly and replaces the parts of the movement, the failure of the movement to work normally is not within the scope of warranty and shall be borne by the user.

9.2 Warranty

Warranty period: the warranty period is one year from the date of shipment of the movement purchased by the user; during the warranty period, the user will not be able to enjoy free service under the following circumstances, and the maintenance cost will be borne by the user:

- A. The operator damages the movement due to improper operation.
- B. Damage caused by the customer's own repair and modification.
- C. Damage caused by the customer's conditions of use not in accordance with the requirements.
- D. Damage caused by natural disasters.